

Topics for week 3

Is $\mathbf{R}^2$ a subset of $\mathbf{R}^3$?

Chapter 2: Vector Spaces

2.3 Subspaces

2.4 Linear independence

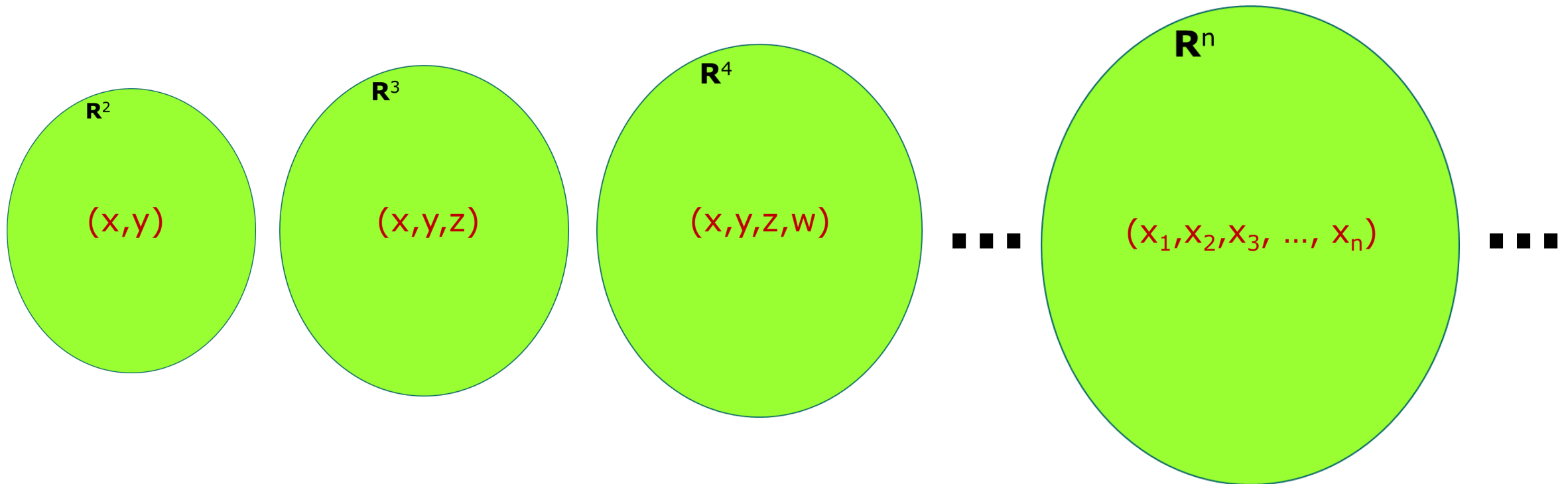
2.5 Basis and Dimension

2.6 Coordinate vectors

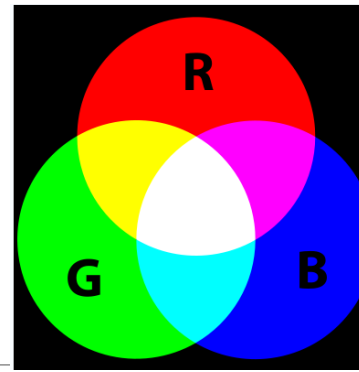
2.7 Projection and Linear approximation

Multiverse

For each dimension n , $\mathbf{R}^n$ is its own “universe”



Color Mixing (An analogy)



Three primary colors: Red, Green, Blue (RGB)

Different color shade combination gives "all" colors

e.g. 20% Red + 45% Green + 30% Blue

this specific linear combination of RGB will produce a particular color

The three primary colors span the entire color space:

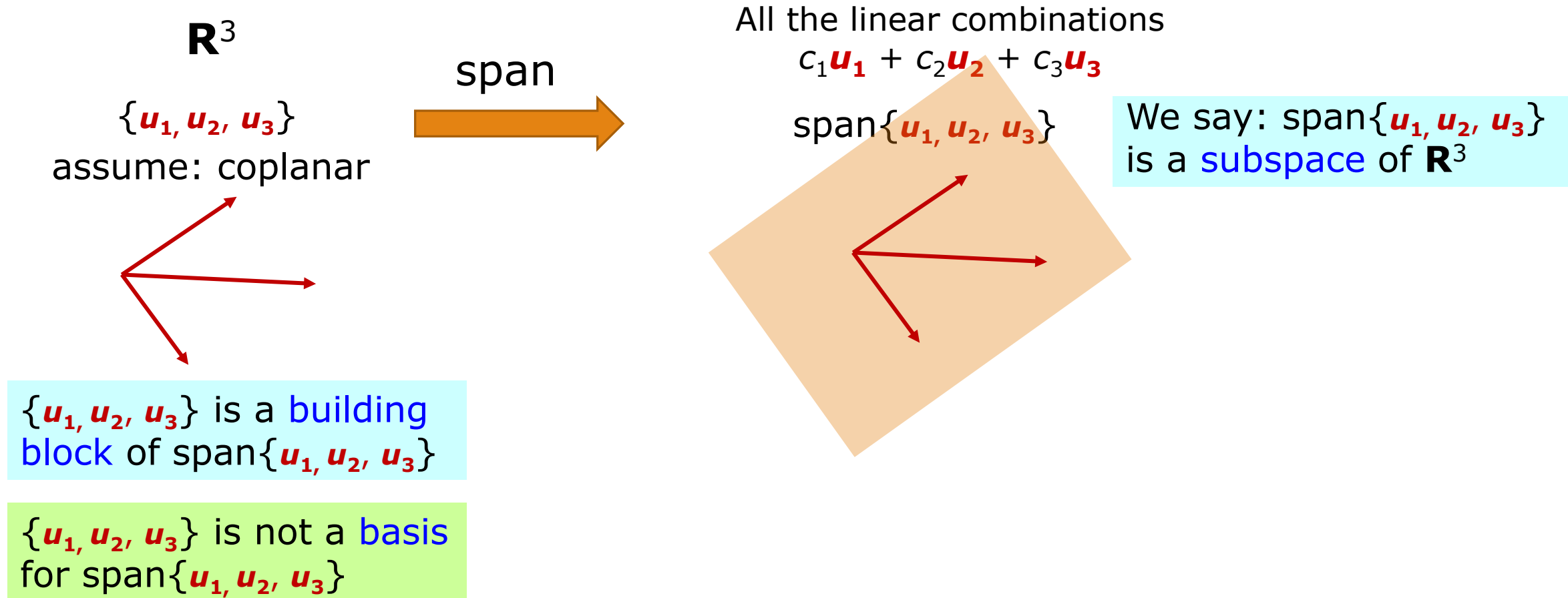
- $\text{span}\{\text{Red}, \text{Green}, \text{Blue}, \text{Yellow}\} = \text{Color space}$
- $\text{span}\{\text{Red}, \text{Green}, \text{Blue}\} = \text{Color space}$

None of the three primary colors are redundant:

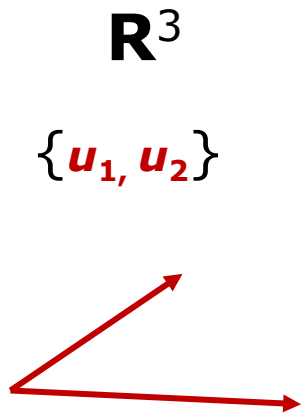
- $\text{Red}, \text{Green}, \text{Blue}, \text{Yellow}$ is not linearly independent
- $\{\text{Red}, \text{Green}, \text{Blue}\}$ is linearly independent

$\{\text{Red}, \text{Green}, \text{Blue}\}$
forms a basis for
the color space

Overview



Overview



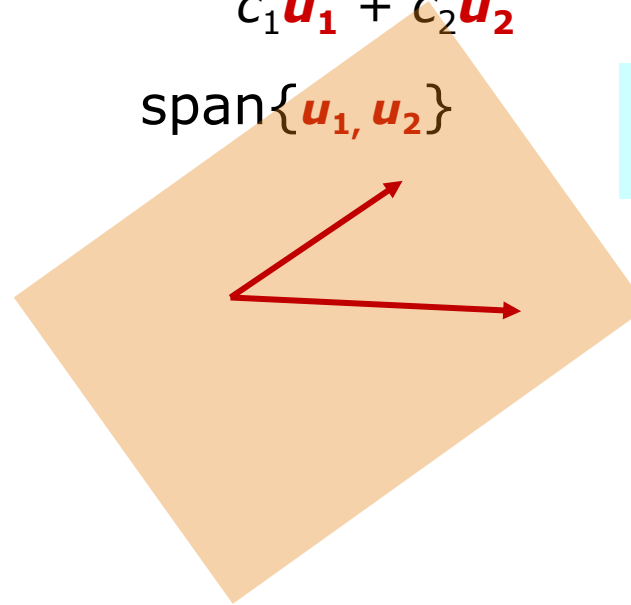
span



All the linear combinations

$$c_1 u_1 + c_2 u_2$$

$\text{span}\{u_1, u_2\}$

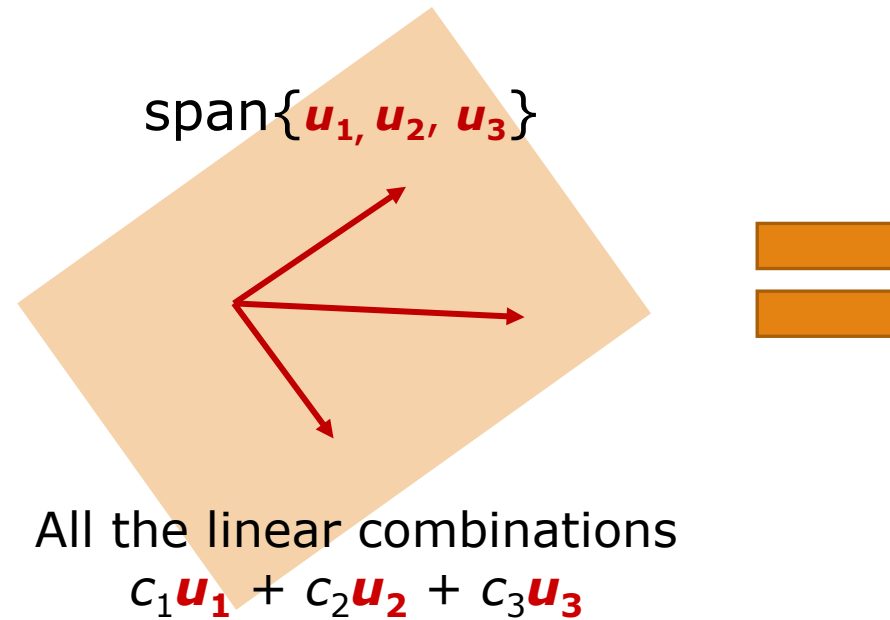


We say: $\text{span}\{u_1, u_2\}$ is a subspace of $\mathbf{R}^3$

$\{u_1, u_2\}$ is a basis for $\text{span}\{u_1, u_2\}$

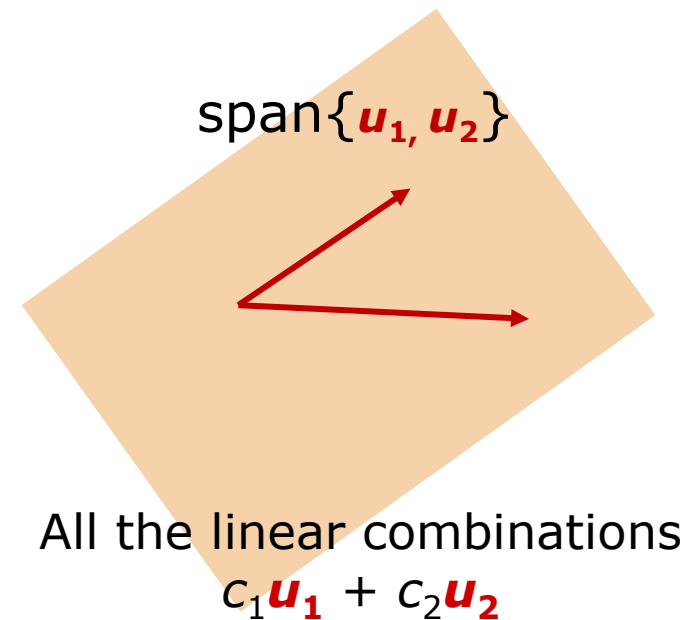
We say the dimension of $\text{span}\{u_1, u_2\}$ is 2

Overview



There is a redundant vector
among $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$

$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a linearly dependent set

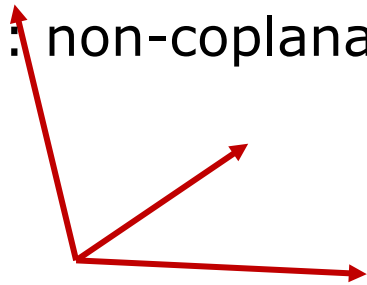


There is no redundant
vector among $\{\mathbf{u}_1, \mathbf{u}_2\}$

$\{\mathbf{u}_1, \mathbf{u}_2\}$ is a linearly independent set

Overview

$\mathbf{R}^3$
 $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$
assume: non-coplanar



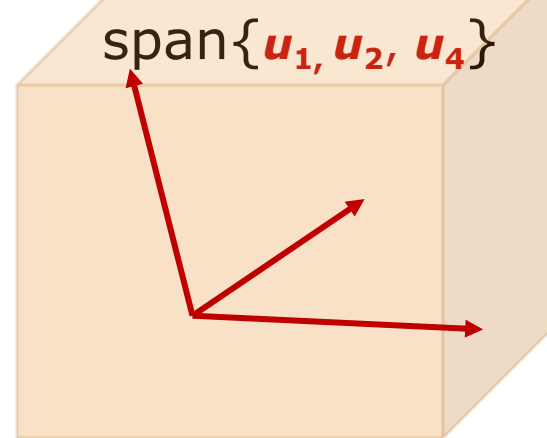
span



All the linear combinations

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_4\mathbf{u}_4$$

$\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$



$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\} = \mathbf{R}^3$$

$\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$ is
again a subspace of $\mathbf{R}^3$

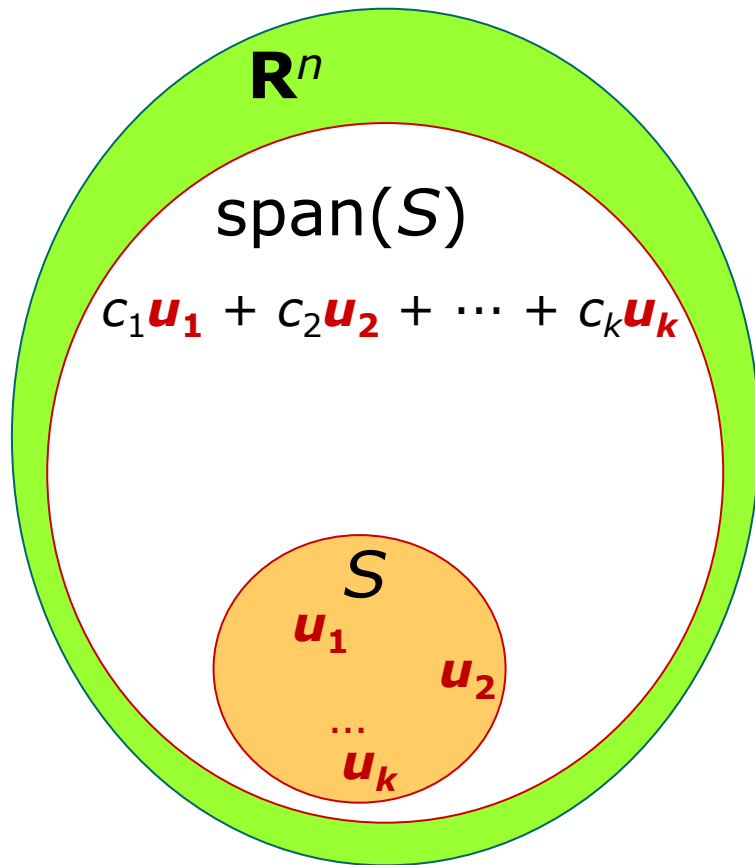
$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$ is a linearly independent set

$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$ is a basis for $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$

$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$ is a basis for $\mathbf{R}^3$

We say the dimension of $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_4\}$ is 3

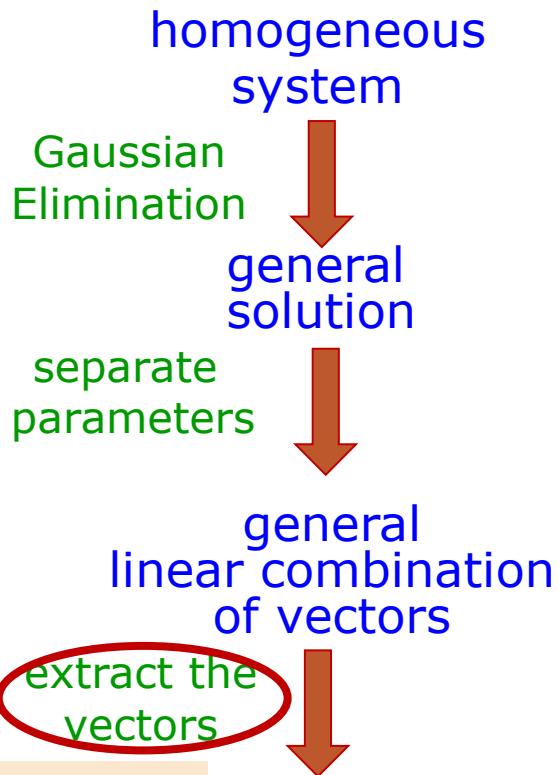
Extending to $\mathbf{R}^n$



- Start with a set S of n -vectors $\{u_1, u_2, \dots, u_k\}$ in $\mathbf{R}^n$
- $\text{span}(S)$ is a **subspace** of $\mathbf{R}^n$
- $\{u_1, u_2, \dots, u_k\}$ is a building block of $\text{span}(S)$
- If $\{u_1, u_2, \dots, u_k\}$ is **linearly independent**, it is a **basis** for $\text{span}(S)$ and the **dimension** of $\text{span}(S)$ is k .

In general,
 $\dim(\text{solution space}) = \text{number of free parameters in the general solution}$

Solution Space of $Ax = 0$



the vectors obtained is always linearly independent

linear span the vectors

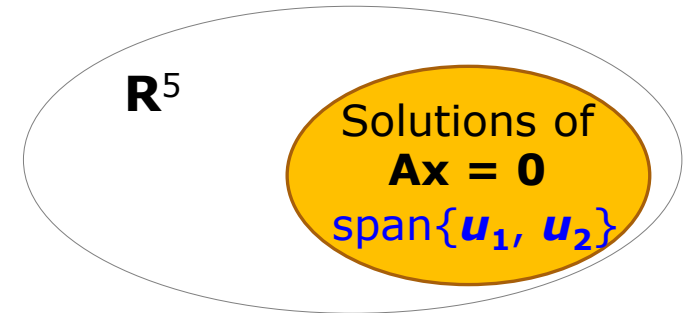
$$\begin{cases} 2v + 2w - x + z = 0 \\ -v - w + 2x - 3y + z = 0 \\ x + y + z = 0 \\ v + w - 2x - z = 0 \end{cases}$$

$$\begin{pmatrix} v \\ w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -s - t \\ s \\ -t \\ 0 \\ t \end{pmatrix}$$

$$s \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

u_1 u_2

$\text{span}\{u_1, u_2\}$ = solution space of $Ax = 0$
 subspace of $\mathbb{R}^5$



basis for solution space = $\{u_1, u_2\}$

$\dim(\text{solution space}) = 2$

G.S of
homog
system

$$\begin{pmatrix} v \\ w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -s-t \\ s \\ -t \\ 0 \\ t \end{pmatrix}$$

G.S of
non-homog
system

$$\begin{pmatrix} v \\ w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -s-t \\ s \\ -t \\ 0 \\ t \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 3 \end{pmatrix}$$

Solution set of linear system

$\mathbf{Ax} = \mathbf{b}$ linear system with n variables

$\mathbf{R}^n$

Solutions of
 $\mathbf{Ax} = \mathbf{b}$

system is homogeneous

$$\mathbf{Ax} = \mathbf{0}$$

The set of solutions is
a "span", and hence a
subspace of $\mathbf{R}^n$

All subspaces of $\mathbf{R}^n$
satisfy closure properties

We call it a **solution space**

Satisfies closure properties

If $\mathbf{u}, \mathbf{v}$ are solutions of $\mathbf{Ax} = \mathbf{0}$,

- $\mathbf{u} + \mathbf{v}$ **also** a solution
- $c\mathbf{u}$ **also** a solution

system is non-homogeneous

$$\mathbf{Ax} = \mathbf{b}$$

The set of solutions is
not a "span", and hence
not a subspace of $\mathbf{R}^n$

We **don't** call it a **solution space**

Does not satisfy closure properties

If $\mathbf{u}, \mathbf{v}$ are solutions of $\mathbf{Ax} = \mathbf{b}$,

- $\mathbf{u} + \mathbf{v}$ **not** a solution
- $c\mathbf{u}$ **not** a solution

Linearly independent set: No redundant vectors in the set

Linearly dependent set: redundant vectors in the set

Example 1 (Linear independence)

Are these sets of vectors in $\mathbf{R}^3$ linearly independent?

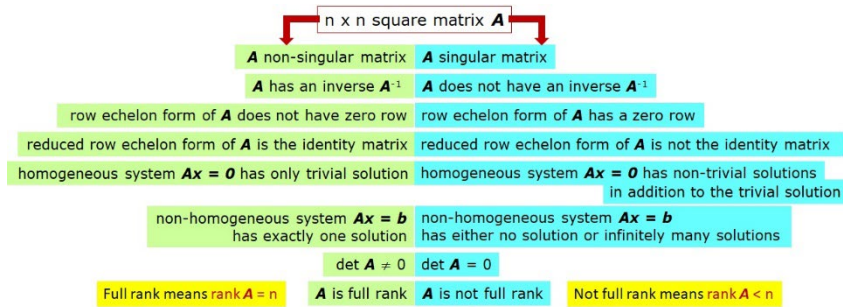
$$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1) \}$$

$$\{ (1, 1, 1), (0, 1, 0), (1, 0, 1) \}$$

$$\{ (1, 1, 1), (-1, -1, -1) \}$$

$$\{ (1, 1, 1), (1, 1, -1) \}$$

$$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1), (1, 0, 1) \}$$



Example 1 (cont.)

$$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1) \}$$

Standard method: $a(1, 1, 1) + b(0, 1, 1) + c(0, 0, 1) = (0, 0, 0)$

Alternative method: $\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 1$ $\begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 1 & 1 & 0 & | & 0 \\ 1 & 1 & 1 & | & 0 \end{pmatrix}$ G.E $\rightarrow$ REF $\rightarrow$ only trivial solution $\rightarrow$ lin indep

$$\{ (1, 1, 1), (0, 1, 0), (1, 0, 1) \}$$

Standard method: $a(1, 1, 1) + b(0, 1, 0) + c(1, 0, 1) = (0, 0, 0)$

Alternative method: $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 0$ $\begin{pmatrix} 1 & 0 & 1 & | & 0 \\ 1 & 1 & 0 & | & 0 \\ 1 & 0 & 1 & | & 0 \end{pmatrix}$ G.E $\rightarrow$ REF $\rightarrow$ non-trivial solution $\rightarrow$ lin dep

Example 1 (cont.)

$\{ (1, 1, 1), (-1, -1, -1) \}$ the set has only two vectors

Standard method

Short-cut method scalar multiples of each other $\rightarrow$ linearly dependent

$\{ (1, 1, 1), (1, 1, -1) \}$ the set has only two vectors

Standard method

Short-cut method not scalar multiples of each other $\rightarrow$ linearly independent

$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1), (1, 0, 1) \}$ the set has more than 3 vectors in $\mathbf{R}^3$

Standard method

Short-cut method more than 3 vectors in $\mathbf{R}^3 \rightarrow$ linearly dependent

Testing Linear dependency

$$\{ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \}$$

Standard Method:

- Form the equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$.

Check whether the homogeneous system has non-trivial solutions for $c_1, c_2, \dots, c_n$.

Alternative Methods: (only work under certain circumstances)

- Use the column vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ to form a matrix $\mathbf{A}$.

If $\mathbf{A}$ is a square matrix, check whether $\det(\mathbf{A}) = 0$.

- If there are only two vectors $\mathbf{v}_1, \mathbf{v}_2$ in the set,
check that $\mathbf{v}_1, \mathbf{v}_2$ are scalar multiple of each other.

- If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbf{R}^m$ and $n > m$,
then the set is linearly dependent.

If Yes,
then linearly dependent
If No,
then linearly independent

If $n < m$, no conclusion

Example 1 revisited (Basis)

Which of the sets of vectors is/are bases for $\mathbf{R}^3$?

$$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1) \}$$

$$\{ (1, 1, 1), (0, 1, 0), (1, 0, 1) \}$$

$$\{ (1, 1, 1), (-1, -1, -1) \}$$

$$\{ (1, 1, 1), (1, 1, -1) \}$$

$$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1), (1, 0, 1) \}$$

Any set of n linearly independent vectors is a basis for $\mathbf{R}^n$

Any set of n linearly independent vectors is a basis for $\mathbf{R}^n$

Example 1 revisited (Basis)

Basis for a
vector space is
NOT UNIQUE

Which of the sets of vectors is/are bases for $\mathbf{R}^3$?

$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1) \}$ 3 vectors, lin indep $\rightarrow$ basis for $\mathbf{R}^3$

$\{ (1, 1, 1), (0, 1, 0), (1, 0, 1) \}$ 3 vectors, lin dep $\rightarrow$ not basis for $\mathbf{R}^3$

$\{ (1, 1, 1), (-1, -1, -1) \}$ Less than 3 vectors $\rightarrow$ not basis for $\mathbf{R}^3$

$\{ (1, 1, 1), (1, 1, -1) \}$ Less than 3 vectors $\rightarrow$ not basis for $\mathbf{R}^3$

$\{ (1, 1, 1), (0, 1, 1), (0, 0, 1), (1, 0, 1) \}$ more than 3 vectors $\rightarrow$ not basis for $\mathbf{R}^3$

$\{ (1, 0, 0), (0, 1, 0), (0, 0, 1) \}$ Standard basis for $\mathbf{R}^3$

Any three non-coplanar vectors form a basis for $\mathbf{R}^3$

G.E. preserves
linear dependency of the columns

Example 2 (Basis for linear span)

$V = \text{span}\{ \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \}$  need to remove the redundant
vectors among $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$

$\text{span}\{(1,2,2,1), (3,6,6,3), (4,9,9,5), (-2,-1,-1,1), (5,8,9,4)\}$

Form a matrix using $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ as its columns

$$\begin{pmatrix} 1 & 3 & 4 & -2 & 5 \\ 2 & 6 & 9 & -1 & 8 \\ 2 & 6 & 9 & -1 & 9 \\ 1 & 3 & 5 & 1 & 4 \end{pmatrix} \xrightarrow{\text{G.E.}} \begin{pmatrix} 1 & 3 & 4 & -2 & 5 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ REF}$$

take columns corresponding
to pivot columns of REF

pivot columns are linearly independent
non-pivot columns are redundant

Basis for the linear span:

$\{(1,2,2,1), \textcolor{red}{(3,6,6,3)}, (4,9,9,5), \textcolor{red}{(-2,-1,-1,1)}, (5,8,9,4)\}$

Dimension of $V = 3$

Subspace in MA1513

- $\mathbf{R}^n$
- $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ where $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are vectors in $\mathbf{R}^n$ subspace of $\mathbf{R}^n$
- Solution space of a homogeneous system $\mathbf{Ax} = \mathbf{0}$ with n variables subspace of $\mathbf{R}^n$
- Row space of an $m \times n$ matrix $\mathbf{A}$ subspace of $\mathbf{R}^n$
- Column space of an $m \times n$ matrix $\mathbf{A}$ subspace of $\mathbf{R}^m$
- Nullspace of an $m \times n$ matrix $\mathbf{A}$ subspace of $\mathbf{R}^n$
- Eigenspace of an $n \times n$ matrix $\mathbf{A}$ (Chapter 3) subspace of $\mathbf{R}^n$

Row Space and Column Space of A

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix} \begin{matrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \\ \mathbf{r}_4 \\ \mathbf{r}_5 \\ \mathbf{r}_6 \end{matrix}$$

$\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{c}_3 \quad \mathbf{c}_4$

row space of $\mathbf{A} = \text{span}\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4, \mathbf{r}_5, \mathbf{r}_6\}$ subspace of $\mathbf{R}^4$

column space of $\mathbf{A} = \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4\}$ subspace of $\mathbf{R}^6$

Nullspace of A

All the vectors $\mathbf{v}$ in $\mathbf{R}^n$ that are “nullified” by $\mathbf{A}$, i.e. $\mathbf{A}\mathbf{v} = \mathbf{0}$.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix}$$

This is same as the solutions of $\mathbf{Ax} = \mathbf{0}$

Nullspace of $\mathbf{A}$ = solution space of $\mathbf{Ax} = \mathbf{0}$
So it is a subspace of $\mathbf{R}^n$

Form the homogeneous system $\mathbf{Ax} = \mathbf{0}$ and solve (using G.E.) as before.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

G.E. preserves
linear dependency of the columns

Basis for Column Space

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{c}_3 \quad \mathbf{c}_4$ REF

column space of $\mathbf{A} = \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4\}$

Basis for column space of $\mathbf{A} =$ columns in $\mathbf{A}$ corresp to pivot columns of REF
 $= \{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$

Dimension of column space $= \#$ pivot columns of REF $= \text{rank}(\mathbf{A}) = 3$

G.E. does not preserve linear dependency of the rows

G.E. preserves the row spaces

Basis for Row Space

$$\dim(\text{row space}) = \dim(\text{column space})$$

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix} \begin{matrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \\ \mathbf{r}_4 \\ \mathbf{r}_5 \\ \mathbf{r}_6 \end{matrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_3 \\ \\ \\ \end{matrix}$$

REF

row space of $\mathbf{A} = \text{span}\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4, \mathbf{r}_5, \mathbf{r}_6\} = \text{row space of REF} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

Basis for row space of $\mathbf{A} = \text{non-zero rows in REF} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

Dimension of row space = # nonzero rows of REF = $\text{rank}(\mathbf{A}) = 3$

Dimension Theorem of Matrix

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = \# \text{ columns of } \mathbf{A}$$

Basis for Nullspace

Nullspace of
matrix $\mathbf{A}$



Solution space of
system $\mathbf{Ax} = \mathbf{0}$

Basis for nullspace



Basis for solution space

Dimension of nullspace



Dimension of solution space



Nullity of $\mathbf{A}$



parameters in general solution

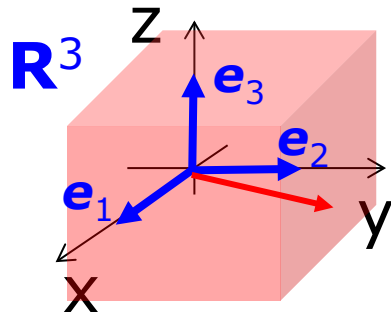
non-pivot columns of REF

Basis as Coordinate System

standard basis

$$S = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

$$\begin{aligned}\mathbf{u} &= (2, 1, -3) \\ &= 2\mathbf{e}_1 + \mathbf{e}_2 - 3\mathbf{e}_3\end{aligned}$$



$$\mathbf{e}_1 = (1, 0, 0)$$

$$\mathbf{e}_2 = (0, 1, 0)$$

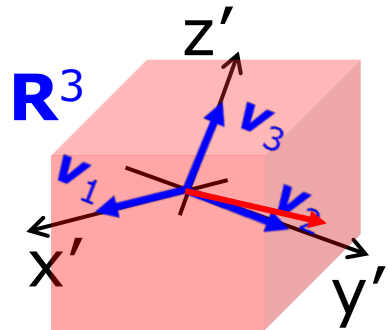
$$\mathbf{e}_3 = (0, 0, 1)$$

$$(\mathbf{u})_S = (2, 1, -3) \quad \text{coordinate vector of } \mathbf{u} \text{ relative to basis } S$$

non-standard basis

$$T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

$$\begin{aligned}\mathbf{u} &= (2, 1, -3) \\ &= 2\mathbf{v}_1 + \mathbf{v}_2 + 0\mathbf{v}_3\end{aligned}$$



$$\mathbf{v}_1 = (1, 0, -1)$$

$$\mathbf{v}_2 = (0, 1, -1)$$

$$\mathbf{v}_3 = (-1, 0, -1)$$

$$(\mathbf{u})_T = (2, 1, 0) \quad \text{coordinate vector of } \mathbf{u} \text{ relative to basis } T$$

Least Squares Solutions

When a linear system $\mathbf{Ax} = \mathbf{b}$ is inconsistent

A least squares solution $\mathbf{x}_0$ of $\mathbf{Ax} = \mathbf{b}$:

- $\mathbf{x}_0$ is the best approximation to a solution of $\mathbf{Ax} = \mathbf{b}$

$$\mathbf{Ax}_0 \underset{\text{closest}}{\approx} \mathbf{b} \quad ||\mathbf{Ax}_0 - \mathbf{b}|| \underset{\text{smallest}}{}$$

How to find the least squares solution $\mathbf{x}_0$ of $\mathbf{Ax} = \mathbf{b}$?

- ❖ Multiply $\mathbf{A}^T$ on both sides of $\mathbf{Ax} = \mathbf{b}$
- ❖ Solve the new system $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$

Example 3 (Least Squares)

$$\begin{array}{l}
 2x + y = 20 \\
 6x + y = 18 \\
 20x + y = 10 \\
 30x + y = 6 \\
 40x + y = 2
 \end{array}
 \quad
 \begin{array}{c}
 \begin{pmatrix} 2 & 1 \\ 6 & 1 \\ 20 & 1 \\ 30 & 1 \\ 40 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 20 \\ 18 \\ 10 \\ 6 \\ 2 \end{pmatrix} \\
 \mathbf{A} \qquad \mathbf{b}
 \end{array}
 \xrightarrow[\mathbf{A}^T]{\begin{pmatrix} 2 & 6 & 20 & 30 & 40 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}}
 \begin{array}{c}
 \begin{pmatrix} 2940 & 98 \\ 98 & 5 \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 608 \\ 56 \end{pmatrix} \\
 \mathbf{A}^T \mathbf{A} \qquad \mathbf{A}^T \mathbf{b}
 \end{array}$$

Find least squares solution $\mathbf{x}_0$ for $\mathbf{Ax} = \mathbf{b} \iff \mathbf{x}_0$ is the actual solution of $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$

What if $\mathbf{A}^T \mathbf{A}$ is singular?

Solve the new system using GE

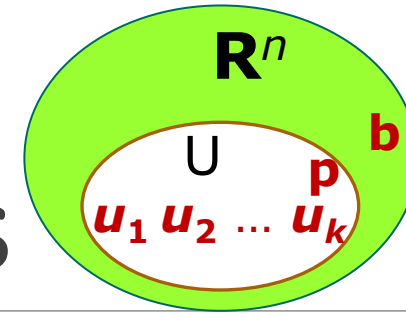
$$\left(\begin{array}{cc|c} 2940 & 98 & 608 \\ 98 & 5 & 56 \end{array} \right)$$

$$\mathbf{x}_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 2940 & 98 \\ 98 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 608 \\ 56 \end{pmatrix} = \begin{pmatrix} -206/637 \\ 268/13 \end{pmatrix}$$

$(\mathbf{A}^T \mathbf{A})^{-1}$

If $\mathbf{x}_0$ is the **least squares solution** of $\mathbf{Ax} = \mathbf{b}$,
 $\mathbf{Ax}_0 =$ **projection** of $\mathbf{b}$ onto column space of $\mathbf{A}$

Projection on Subspaces

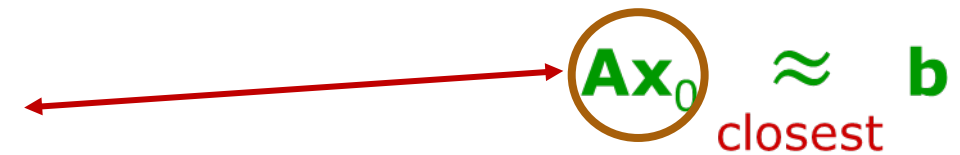


Given a subspace U of $\mathbf{R}^n$ and a vector $\mathbf{b}$ not in U

Find a vector $\mathbf{p}$ in U “nearest” to $\mathbf{b}$

How to find this projection of $\mathbf{b}$?

This vector $\mathbf{p}$ is called
the **projection** of $\mathbf{b}$ onto a subspace U



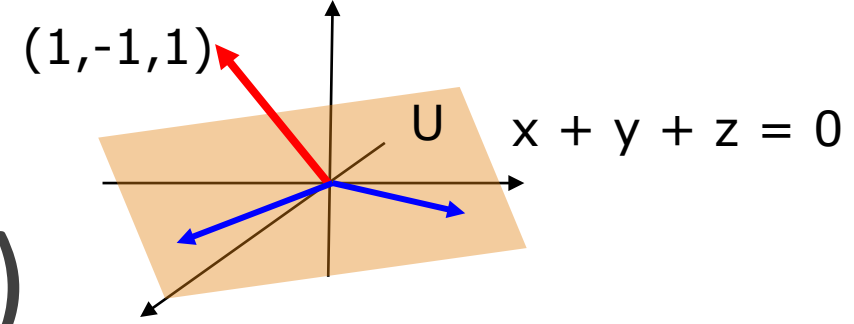
- Find **any** basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ for U
- Form matrix $\mathbf{A}$ using $\mathbf{u}_1, \dots, \mathbf{u}_k$ as column vectors
- Consider the linear system $\mathbf{Ax} = \mathbf{b}$
- Solve the system $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$, say $\mathbf{x}_0$
- **projection** $\mathbf{p} = \mathbf{Ax}_0$

$$\rightarrow \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = U$$

$$\rightarrow \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} \\ = \text{column space of } \mathbf{A}$$

least squares solution of $\mathbf{Ax} = \mathbf{b}$

If $\mathbf{x}_0$ is the **least squares solution** of $\mathbf{Ax} = \mathbf{b}$,
 $\mathbf{Ax}_0 =$ **projection** of $\mathbf{b}$ onto column space of $\mathbf{A}$



Example 4 (Projection)

Find the **projection** of $(1, -1, 1)$ onto the plane $U: x + y + z = 0$ in $\mathbf{R}^3$

$$U = \text{span}\{ \text{2 non-parallel vectors on the plane} \} = \text{span}\{ (1, -1, 0), (1, 0, -1) \}$$

Form matrix $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}$ and column vector $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

Find the least squares solution of $\mathbf{Ax} = \mathbf{b}$

Solve $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$ $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \longrightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4/3 \\ -2/3 \end{pmatrix}$

Projection of $(1, -1, 1)$ on U : $\begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 4/3 \\ -2/3 \end{pmatrix} = \begin{pmatrix} 2/3 \\ -4/3 \\ 2/3 \end{pmatrix}$